

# Ali Peivand

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## Education

### Master's Degree

|                 |                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------|
| Dates           | 2021 – 2024                                                                             |
| Institution     | Isfahan University of Technology, Iran                                                  |
| Field of study  | Electrical Engineering (Majoring in <i>Power Systems</i> )                              |
| GPA             | 16.78/20                                                                                |
| Title of thesis | Wind Curtailment Mitigation with Energy Storage Systems (Electric vehicles & batteries) |
| Thesis mark     | 20 out of 20                                                                            |

|                 | Course                                       | Mark (out of 20) |
|-----------------|----------------------------------------------|------------------|
| Notable courses | • Energy Distribution                        | 19               |
|                 | • Renewable Energy                           | 19.45            |
|                 | • Special Topics1(Power System Optimization) | 18               |
|                 | • Special Topics2(Smart Grids)               | 17.5             |
|                 | • Dynamic of Power System 1                  | 16               |

## Publications

|                       |                                                                                                                                                                                                                                                                                                                         |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Published             | • Ali Peivand, "A PPO-GAN-Enabled Moving Target Defense for Cyber-Resilient and Cost-Aware Power System Scheduling", <i>IEEE Transactions on Industrial Informatics</i> , <a href="https://doi.org/10.1109/TII.2026.3675324">https://doi.org/10.1109/TII.2026.3675324</a> .                                             |
| Published             | • Ali Peivand, "Enhancing Power Grid Cybersecurity Against FDI Attacks via Deep Q-Network-Based Moving Target Defense", <i>Reliability Engineering &amp; System Safety</i> , <a href="https://doi.org/10.1016/j.ress.2026.112390">https://doi.org/10.1016/j.ress.2026.112390</a> .                                      |
| Published             | • Ali Peivand, "Wind curtailment mitigation in presence of battery energy storage and electric vehicle: A comprehensive multi objective decision-support framework." <i>Journal of Cleaner Production</i> , <a href="https://doi.org/10.1016/j.jclepro.2023.137215">https://doi.org/10.1016/j.jclepro.2023.137215</a> . |
| Published             | • Ali Peivand, "Accelerating Optimal Scheduling Prediction in Power System: A Multi-Faceted GAN-Assisted Prediction Framework." <i>Renewable Energy</i> (2024). <a href="https://doi.org/10.1016/j.renene.2024.120830">https://doi.org/10.1016/j.renene.2024.120830</a> .                                               |
| Published             | • A. Peivand, "Intelligent parking lot power management: Augmented epsilon-constraint concept with correlation analysis," <i>IET Renewable Power Generation</i> , <a href="https://doi.org/10.1049/rpg2.13143">https://doi.org/10.1049/rpg2.13143</a> .                                                                 |
| Accepted (In Persian) | • Ali Peivand, "Optimal decision-making employment using a risk-based multi-stage model in the multi-objective optimal scheduling process of micro-energy grid based on CCHP with using augmented epsilon-constrained concept" <i>Journal of Modeling in Engineering</i> (2024).                                        |
| Under Review          | • Ali Peivand, "A Hierarchical Deep Generative–Optimization Framework for Real-Time Economic Scheduling of Wind-Integrated Power Systems", <i>Journal of Computers and Electrical Engineering</i> (2026).                                                                                                               |
| Under Review          | • Ali Peivand, "A Wavelet-Driven Bi-Level Optimization Framework in Power Systems: Deep-Aided Optimization with CVaR-Based Risk Management", <i>Journal of Electric Power Systems Research</i> (2025).                                                                                                                  |

## Honours & Awards

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- 2025 – 2029      Full Doctoral Scholarship, New Mexico State University (NMSU), USA — 2025–2029  
                     ➤ *Awarded a full PhD funding package including tuition waiver and university scholarship.*
- 2025 – 2029      Graduate Research Assistantship Award, New Mexico State University (NMSU), USA — 2025–2029  
                     ➤ *Awarded by supervisor with annual stipend of \$33,000 to support doctoral research.*
- 2026              Academic Service & Peer Review  
                     ➤ *Ad hoc Reviewer for IEEE Transactions on Industrial Cyber-Physical Systems, IEEE Industrial Electronics Society, 2026. Reviewed manuscripts in the areas of cyber-physical systems, power SCADA security, optimization, and resilient distributed energy resources.*
- 2021 – 2024      Awarded tuition fee waiving scholarship for my master's studies, IUT
- 2016 – 2021      Awarded tuition fee waiving scholarship for my bachelor's studies, SUT

## Academic Projects Completed

### Optimization & Power System Scheduling:

- Developed and implemented Particle Swarm Optimization (PSO) and formulated AC/DC Optimal Power Flow (OPF) and Unit Commitment (UC) problems using NLP and MINLP frameworks.

### Deep Learning for Renewable Forecasting:

- Built deep neural network (DNN) models in MATLAB and a Bi-LSTM-based wind power prediction model in Python.

### Probabilistic Power System Modeling:

- Simulated an 83-bus solar–wind-integrated power grid using Monte Carlo Simulation (MCS) to capture stochastic behavior under varying operating conditions.

### Machine Learning & Data Analytics:

- Implemented regression models (KNN, SVM, Random Forest, Decision Tree) and applied K-means clustering to analyze electricity price datasets.

### Optimization & Power System Scheduling:

- Developed and implemented Particle Swarm Optimization (PSO) and formulated AC/DC Optimal Power Flow (OPF) and Unit Commitment (UC) problems using NLP and MINLP frameworks.

### Deep Learning for Renewable Forecasting:

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### Probabilistic Power System Modeling:

- Simulated an 83-bus solar–wind-integrated power grid using Monte Carlo Simulation (MCS) to capture stochastic behavior under varying operating conditions.

### Machine Learning & Data Analytics:

- Implemented regression models (KNN, SVM, Random Forest, Decision Tree) and applied K-means clustering to analyze electricity price datasets.

## Research Interests

### Control and Optimization of Power Systems

AI-driven control, optimal scheduling, AC/DC optimal power flow, unit commitment, stochastic and robust optimization, bi-level optimization, CVaR-based risk management, and real-time decision-making in renewable-rich power systems.

### Cybersecurity and Cyber-Resilient Smart Grids

### Smart Grids and Microgrids

Operation and energy management of smart grids, microgrids, renewable energy resources, electric vehicles, battery energy storage systems, CCHP-based energy hubs, and intelligent parking lots under uncertainty and flexibility constraints.

### Advanced AI and Deep Learning Applications

Cyber-resilient operation of power cyber-physical systems, SCADA security, False Data Injection attacks, Denial-of-Service attacks, Moving Target Defense, secure scheduling, and resilient distributed energy resource management.

Deep reinforcement learning, PPO, Deep Q-Networks, GANs, Bi-LSTM, transformers, physics-informed machine learning, and deep generative optimization for forecasting, control, cybersecurity, and optimization of smart energy systems.

## Work Experience

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### Python, MATLAB, and GAMS Instructor (Part-Time) | Kerman, Iran

June 2024 – August 2024

- **Teaching Python:** Instructed students at various levels, from beginners to advanced, on Python programming, Machine Learning & Deep Learning.
- **MATLAB Instruction:** Educated advanced students on MATLAB, focusing on both the scripting environment and Simulink for mathematical optimization, simulation, and modeling. Covered metaheuristic algorithms such as Genetic Algorithms and applications in simulating power plants.
- **GAMS Training:** Provided in-depth instruction on GAMS, ideal for tackling complex and large-scale optimization problems. Focused on its application in mathematical optimization within power systems and other engineering fields.
- **Power System Analysis:** Integrated Python, MATLAB, and GAMS teaching with power system analysis concepts, including optimization and simulation using libraries and tools like PYOMO and PyPSA.

## Certificates Awarded

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- [K-Means Clustering in Python](#), University of London; 2024
- [Pandas course on Kaggle](#); 2024
- [Computer Vision on Kaggle](#); 2025
- [Deep Learning on Kaggle](#); 2025
- PYTHON professional license issued by Ilia Training Center, Kerman, Iran; 2022
- MATLAB professional license issued by Ilia Training Center, Kerman, Iran; 2017

## Skills

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### Computer Skills

- *Programming Languages:* Python, MATLAB (Simulink), TensorFlow, Keras, PyTorch, Pandapower, Pandas, Numpy, PYomo, PyPSA, CVXPY, Scikit-learn, Optuna
- R
- GAMS
- *Professional Software:* Power Factory (DIgSILENT)
- *General Software:* Microsoft Office (Viso, Power BI), Git, SQL

### Language Skills

- Persian (Native)
- English (Advanced); IELTS Overall Score: 7 (L: 7.5, R: 6.5, S: 7.5, W: 6.5)

## References

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**Dr. Seyyed Mostafa Nosratabadi, Associate Professor**

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**Dr. Hamid Reza Abdolmohammadi, Associate Professor**

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**Dr. Ehsan Azad-Farsani, Associate Professor**

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